DEPARTMENT EVENT TICKET BOOKING

School of Computing

Bachelor of Technology
in
Computer Science & Engineering

By

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Full Stack Application Development
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BONAFIDE CERTIFICATE

It is certified that the work contained in the Full Stack Application Development report titled “Department Event Ticket Booking” by S.Venkata Naga Sai (VTU26083) has been carried out in the Winter Semester of the Academic Year 2025–2026 (WS2526).

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ABSTRACT

The Ticket Booking of an Internal Department Event is a web-based application developed using React JS to simplify the process of booking tickets for department events such as technical fests, seminars, workshops, and cultural programs. The system allows students and faculty members to view complete event details including event name, department name, date, time, venue, ticket price, and available seats. Users can easily book tickets online by entering their personal details such as name, email, department, and required number of tickets. The application performs proper input validation, prevents overbooking, and displays confirmation messages after successful booking. It also updates the available ticket count dynamically and gen-erates a booking summary with total amount. The project uses React concepts such as functional components, JSX, useState hook, event handling, and conditional ren-dering to create an interactive and user-friendly interface.

Keywords: React JS, Ticket Booking, Department Event, Web Application

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LIST OF ACRONYMS AND ABBREVIATIONS

API	Application Programming Interface
CRUD	Create, Read, Update, Delete
CSS	Cascading Style Sheets
DOM	Document Object Model
HTML	HyperText Markup Language
JS	JavaScript
JSON	JavaScript Object Notation
SPA	Single Page Application
UI	User Interface
UX	User Experience

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Chapter 1

INTRODUCTION

1.1 Introduction

The Ticket Booking System for an Internal Department Event is designed to provide an efficient and reliable solution for managing event registrations within an organization or academic department. Traditional ticket booking methods are often time-consuming and prone to errors, especially when handling many participants. This system offers a digital platform where users can view event details, check avail-ability, and book tickets securely, while administrators can manage events and mon-itor bookings in real time.

1.2 Aim of the Project

The aim of this project is to develop a user-friendly web-based system for book-ing tickets for internal department events. It focuses on automating the registration process to reduce manual effort and errors. The system also aims to provide a secure and efficient platform for managing event details and user bookings. Additionally, it enhances user experience by offering easy navigation, real-time updates, and reliable data management.

1.3 Scope of the Project

The scope of the Ticket Booking System for an Internal Department Event in-cludes developing a web-based platform that allows users to view event details, check ticket availability, and book tickets online. It covers features such as secure user authentication, event creation, ticket management, and booking tracking by administrators. The system is intended for internal use within a department or organization, aiming to improve efficiency, reduce manual work, and ensure accurate and orga-nized event management.(

1.4 Framework

The system framework consists of layers such as the user interface, the application logic, and the database(Fig.1.1).

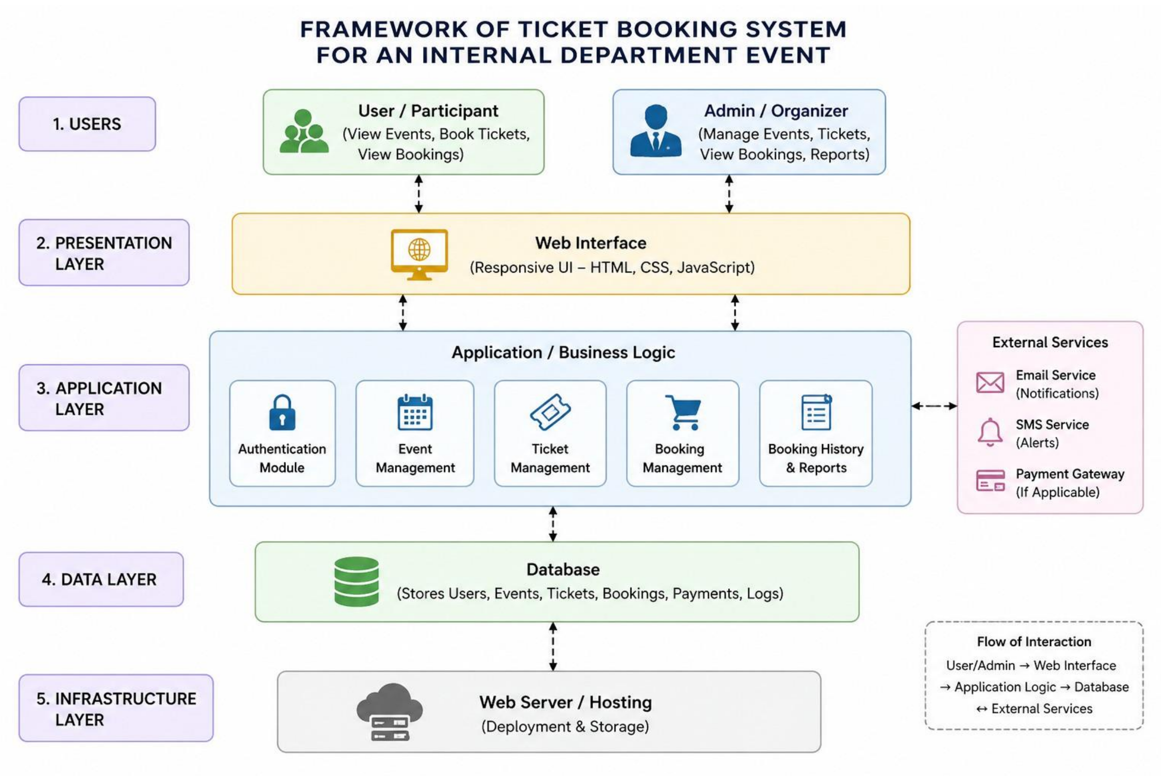

Figure 1.1: Framework

Chapter 2

SURVEY OF SIMILAR APPLICATIONS IN THE MARKET

Several ticket booking and event management applications are widely used in the market, such as BookMyShow, Eventbrite, Ticketmaster, and Paytm Insider. These platforms provide essential features including real-time ticket booking, secure online payment integration, event discovery, and user-friendly interfaces that enhance the overall user experience.

In addition to these, platforms like Eventzilla and Arlo Event Management offer advanced functionalities such as customizable event pages, attendee management, automated communication, and detailed reporting tools for organizers. These fea-tures help in efficiently managing large-scale events and tracking participant engage-ment.

Furthermore, systems like Booking Ninjas provide real-time tracking, automated registration processes, and centralized dashboards, which significantly improve operational efficiency and reduce manual effort. Many of these platforms also support multi-device access, enabling users to book tickets through web and mobile applications seamlessly.

Most existing applications are primarily designed for large-scale public events such as concerts, movies, and conferences. They commonly include features like seat selection, QR-based e-ticketing, multiple payment options, and analytics tools that support data-driven decision-making. These systems are highly scalable and capable of handling a large number of users simultaneously.

However, these platforms often lack specific support for internal departmental or institutional events, where requirements such as restricted access, simplified book-ing, and internal communication are essential. They may also include complex fea-tures that are unnecessary for smaller-scale events, making them less suitable for internal use.

This gap highlights the need for a dedicated system tailored for internal event management and ticket booking. The proposed system focuses on simplicity, controlled access, and efficient management, making it more suitable for academic departments or organizational environments.

Chapter 3

FRAMEWORK DESCRIPTION

3.1 Frontend Implementation

3.1.1 Environment Setup

The frontend development environment is set up using modern web technologies such as HTML, CSS, and JavaScript. React.js is used to build a dynamic and responsive user interface. Development tools such as Visual Studio Code and web browsers are used for coding and testing. Required libraries and dependencies are installed using Node Package Manager (npm) to enhance functionality and design.

3.1.2 Structure of the Project

The project follows a modular structure to ensure scalability, maintainability, and separation of concerns:

- **/public:** Contains static assets such as index.html, logos, and images used across the application.
- **/src/api:** Includes configuration files (e.g., axios.js) for handling communication with the backend server.

- **/src/components:** Contains reusable UI components such as Navbar, Sidebar, Event Cards, and Ticket Forms.
- **/src/context:** Manages global state using Context API (e.g., AuthContext.js) for authentication and session handling.
- **/src/pages:** Includes the main application pages:
- **/src/pages:** Includes the main application pages:
  - HomePage.jsx – Displays available events and general information
  - EventDetailsPage.jsx – Shows detailed information about selected events
  - AdminDashboardPage.jsx – Admin panel for managing events and bookings
  - AdminLoginPage.jsx – Handles administrator authentication
  - MyTicketsPage.jsx – Displays user booking history and tickets
- **/src/utils:** Contains helper functions such as validation and data formatting.
- **App.jsx:** The root component that manages routing, navigation, and overall layout of the application.

3.1.3 Execution Procedure

The frontend application is executed using a local development server. The project is started using the following commands:

```
npm install
npm start
```

Once the server is running, the application can be accessed through a web browser, allowing users to interact with the system, view events, and book tickets.

3.2 Backend Implementation

The backend server is developed using Spring Boot. It handles client requests, processes business logic, and communicates with the database. The server can be started using:

```
mvn spring-boot:run
```

Once running, it manages user authentication, event operations, and booking processes. APIs are tested using tools like Postman to ensure proper functionality(Fig.3.1).

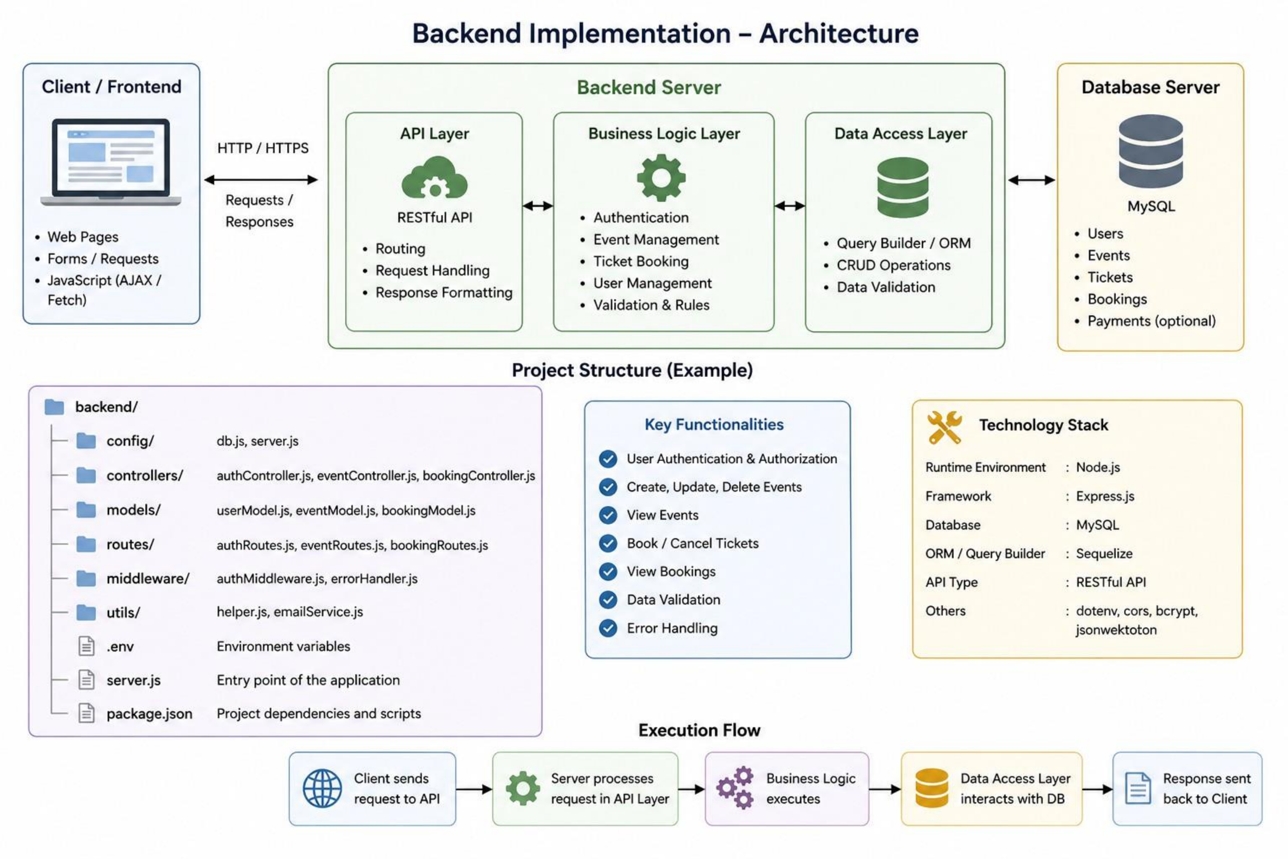

Figure 3.1: Backend Implementation

3.3 Database Implementation

3.3.1 Database Configuration

The database is configured using MySQL, a relational database management system that provides reliable and efficient data storage. It is used to store and manage all application data, including user information, event details, ticket availability, and booking records.

The configuration involves creating the required database schema, defining tables with appropriate fields, and establishing relationships using primary and foreign keys. Proper indexing is applied to improve query performance and ensure faster data retrieval.

Connectivity between the backend and the database is established using JDBC, allowing the application to perform operations such as data insertion, retrieval, updating, and deletion. Security measures such as password encryption and controlled access permissions are implemented to protect sensitive user data.

Overall, the database configuration ensures data integrity, consistency, and efficient handling of large volumes of event and booking information.

3.3.2 Schema Structure

The database schema consists of structured tables with proper relationships using primary and foreign keys. It includes entities such as Users, Events, Tickets, Bookings, and Payments to maintain data integrity and consistency.

3.3.3 Tables Created

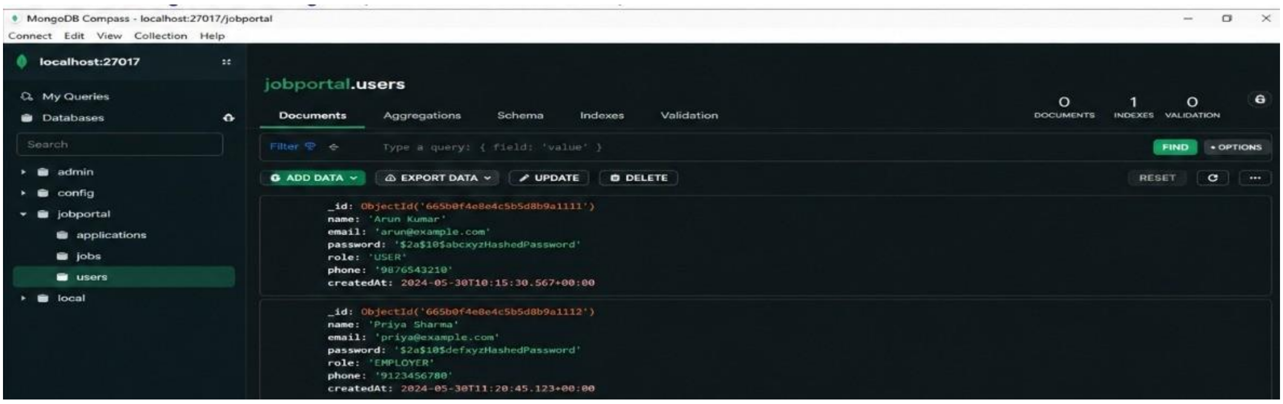

Figure 3.2: Data configuration

This image 3.2 illustration demonstrates how MongoDB is used as a NoSQL database to manage and store user information in a job portal system, supporting authentication, role-based access, and data organization.

3.4 System Technology Stack Overview

The following table summarizes the technologies used in the development of the system:

Table 3.1: System Technology Stack

Component	Technology
Frontend	React.js, HTML5, CSS3
Backend	Spring Boot, Spring Security
Database	MySQL
API Communication	Axios
Tools	Git, Postman, VS Code

Table 3.1 presents the overall technology stack used in the development of the system. The application follows a modern full-stack architecture where different technologies are integrated to ensure scalability, security, and efficient performance.

Chapter 4

METHODOLOGIES

4.1 Project Architecture

The project follows a layered architecture consisting of three main components: frontend, backend, and database. This architecture ensures a clear separation of concerns, making the system more organized, maintainable, and scalable.

The frontend layer is responsible for handling user interactions and presenting information through a responsive user interface. It allows users to view event details, select tickets, and perform booking operations easily.

The backend layer manages the core application logic, including user authentication, event management, ticket processing, and validation. It acts as a bridge between the frontend and the database, ensuring secure data handling and proper execution of operations.

The database layer is used to store and manage all application data such as user information, event details, ticket availability, and booking records. It ensures data consistency, integrity, and efficient retrieval.

Overall, this layered architecture enhances system performance, security, and scalability, making it suitable for handling multiple users and real-time booking operations[fig 4.1].

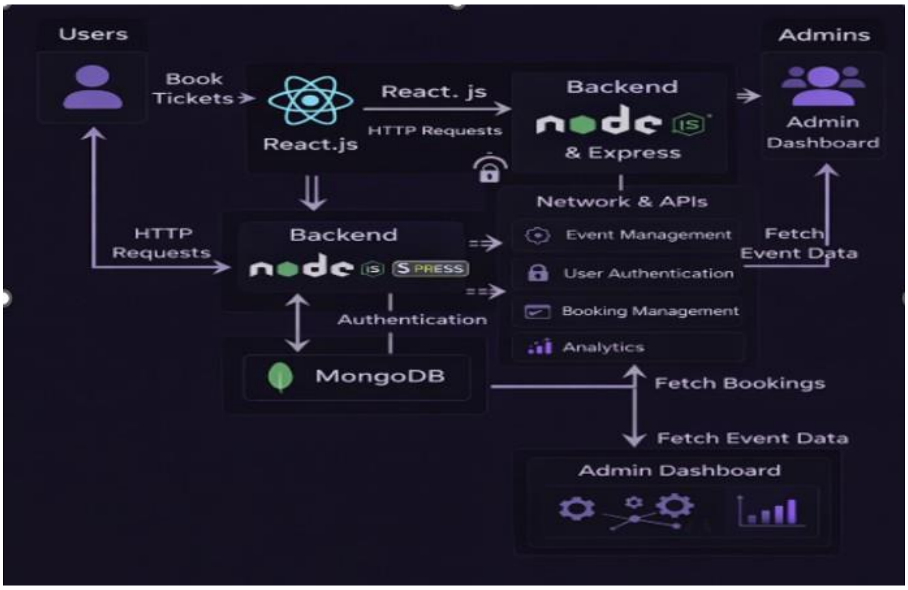

Figure 4.1: System Architecture

4.1.1 Workflow

A Data Flow Diagram (DFD) is a graphical representation that shows how data moves through a system. It explains how information is input, processed, stored, and output within a system. DFD helps in understanding the overall working of a system in a simple and visual way[fig4.2].

Figure 4.2: Workflow

4.2 DataFlow Diagram

The Data Flow Diagram (DFD) represents the flow of data within the system and illustrates how information is processed between different components.

Initially, the user provides input through the frontend interface, such as selecting an event or entering booking details. This input is then sent to the backend server, where it is processed and validated according to the system logic.

The backend communicates with the database to store or retrieve relevant information, such as checking ticket availability or saving booking records. After processing, the backend sends the response back to the frontend(Fig.4.3).

Finally, the system displays the output to the user, such as booking confirmation, updated ticket availability, or error messages if any issue occurs. This continuous flow ensures smooth communication between all components and efficient system functionality.

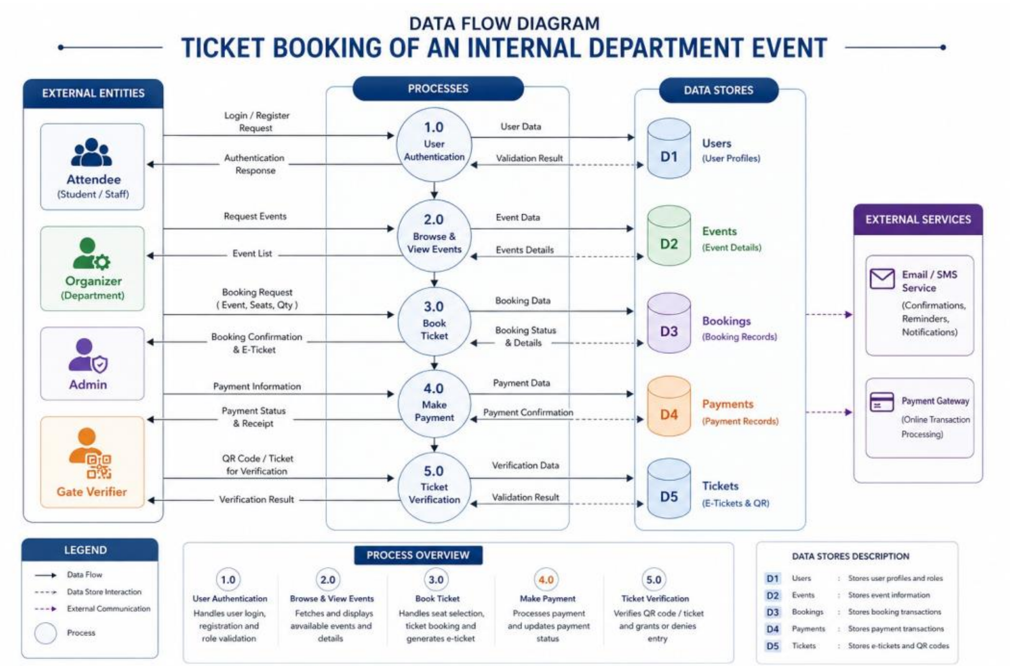

Figure 4.3: DataFlowDiagram

4.3 Module 1:

User Management: This module handles user registration, login, and authentication. It ensures that only authorized users can access the system and perform booking operations.

4.4 Module 2:

Event Management: This module allows administrators to create, update, and manage event details such as date, time, and ticket availability.

4.5 Module 3:

Ticket Booking: This module enables users to book tickets, check availability, and receive booking confirmation. It also prevents overbooking through proper vali-dation.

Advantages of this Project The system simplifies ticket booking, reduces manual work, and improves efficiency. It ensures accurate data management and provides a user-friendly interface for both users and administrators.

a) Time Saving: Automates the booking process, reducing manual effort.

b) Error Reduction: Minimizes human errors and prevents overbooking.

c)Easy Management: Helps administrators manage events and bookings efficiently.

Disadvantages The system depends on internet connectivity and may face technical issues if not maintained properly. Initial setup and development may require time and technical knowledge.

Chapter 5

RESULTS AND DISCUSSIONS

The Ticket Booking System for an Internal Department Event is developed as a comprehensive platform that streamlines event management and ticket booking processes within an organization. The system integrates multiple functional modules to provide a seamless and efficient user experience for both users and administrators.

5.1 System Modules and Features

The implemented system consists of the following key modules:

- **Landing Page Interface:** Provides an overview of available events and navigation options for users.
- **User Authentication Module:** Handles user registration, login, and secure authentication.
- **Event Listing and Management Module:** Displays event details and allows administrators to create, update, or delete events.
- **Ticket Booking System:** Enables users to book tickets by selecting events and entering required details.

- **Seat Selection Module:** Allows users to choose seats visually and ensures availability before booking.
- **Payment Processing Module:** Manages payment transactions and confirms successful bookings (if applicable).
- **E-Ticket and QR Code Generation:** Generates digital tickets with QR codes for easy verification.
- **User Dashboard (My Tickets):** Allows users to view their booking history and download tickets.
- **Admin Dashboard:** Provides administrative control for managing events, users, and bookings.
- **Notification System:** Sends alerts via email or SMS for booking confirmations and updates.
- **Reporting and Analytics Module:** Generates reports and insights related to bookings and events.
- **Database Layer:** Stores all system data including users, events, tickets, and transactions.

5.2 System Performance and Discussion

The system follows a modular and layered architecture that ensures scalability, secure communication, and efficient handling of event-related operations. It enables smooth interaction between users and administrators while maintaining data integrity and real-time updates throughout the booking process.

The performance of the system has been evaluated based on responsiveness, data handling capability, and user interaction efficiency. The application provides quick response times during event browsing, ticket booking, and data retrieval operations. The integration between the frontend and backend ensures seamless data flow with-out noticeable delays.

The system effectively manages multiple user requests and maintains consistency in database operations. Proper validation mechanisms are implemented to prevent incorrect data entry and ensure reliability. The use of structured queries for storing and retrieving data enhances performance and accuracy.

In terms of usability, the interface is designed to be simple and intuitive, allowing users to easily navigate through different features such as viewing events, booking tickets, and accessing booking details. The administrator module provides efficient control over event management, including adding, updating, and monitoring event information.

The system is flexible and can be further enhanced by integrating advanced features such as online payment gateways and notification services. It also provides a strong foundation for future expansion to support larger scale event management systems.

The results demonstrate that the system significantly reduces manual effort, minimizes human errors, and improves overall operational efficiency. Automated processes such as data storage, retrieval, and validation contribute to better system performance. Overall, the application provides a reliable, scalable, and user-friendly solution for managing internal department events.

5.3 System Interface Visualizations

5.3.1 Landing Page Interface

The landing page (Fig. 5.1) provides an overview of the system and introduces users to available events. It presents a clean and interactive interface

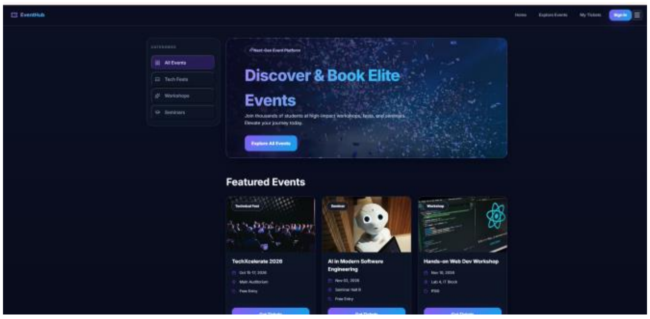

Figure 5.1: Landing Page Interface:

5.3.2 User Authentication:

The user authentication module (Fig. 5.2) allows users to securely log in using valid credentials. It ensures authorized access and maintains data security through proper validation mechanisms.

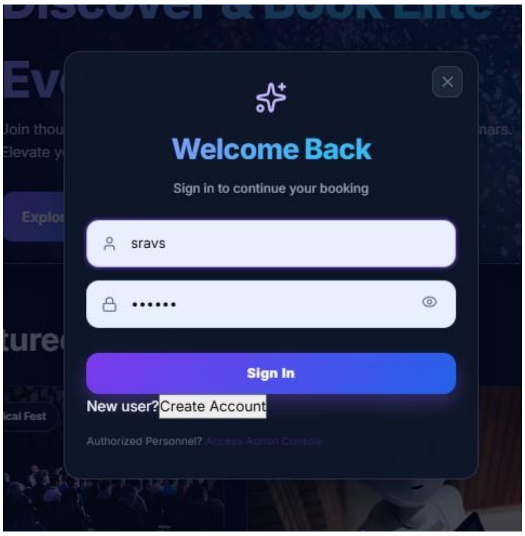

Figure 5.2: User Authentication

5.3.3 Event Listing Management

The event listing module (Fig. 5.3) displays all available events with details such as title, description, and booking options. It helps users efficiently explore and select events of their interest.

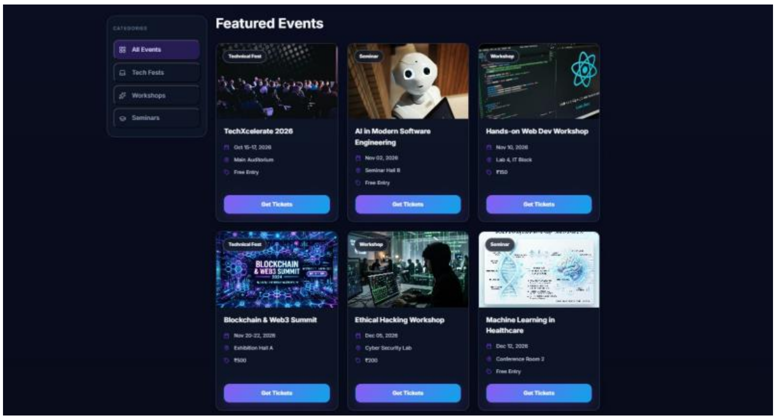

Figure 5.3: Event Listing Management

5.3.4 Ticket Booking System

The ticket booking system (Fig. 5.4) allows users to select seats and proceed with the booking process. It ensures a smooth and error-free experience through a structured and guided workflow.

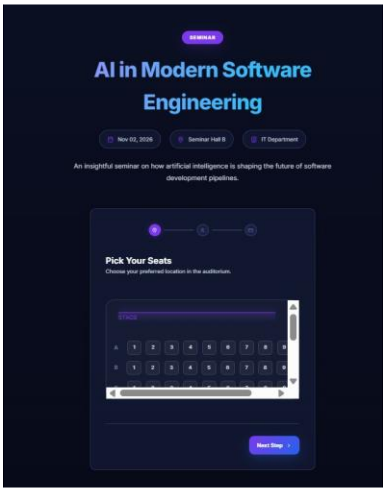

Figure 5.4: Ticket Booking System

5.3.5 Booking Details

The booking details module (Fig. 5.5) collects necessary user information re-quired to complete the reservation. It captures details such as name, contact in-formation, and booking preferences. The module ensures that all mandatory fields are properly filled before submission. Validation checks are applied to maintain ac-curacy and avoid incorrect entries. This helps in ensuring a smooth and reliable booking process.

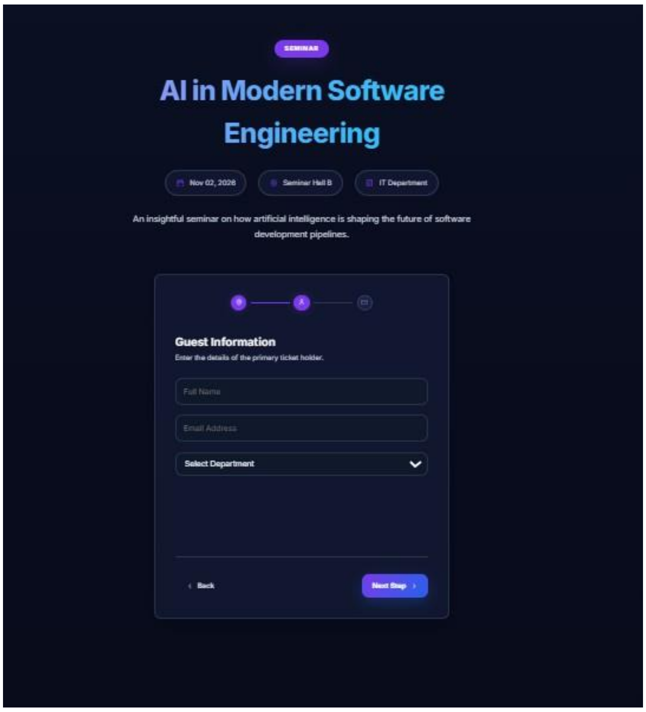

Figure 5.5: Booking Details

5.3.6 Payment Processing

The payment processing module (Fig. 5.6) displays the booking summary and allows users to confirm their reservation. It ensures secure and reliable transaction handling.

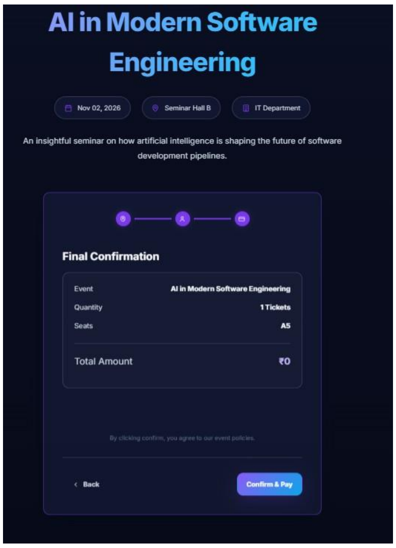

Figure 5.6: Payment Processing

5.3.7 Ticket Generation

The ticket generation module (Fig. 5.7) produces a digital ticket after successful booking. The generated ticket contains all relevant information such as event details and booking confirmation.It Provides users with a clear and verifiable proof of their reservation.

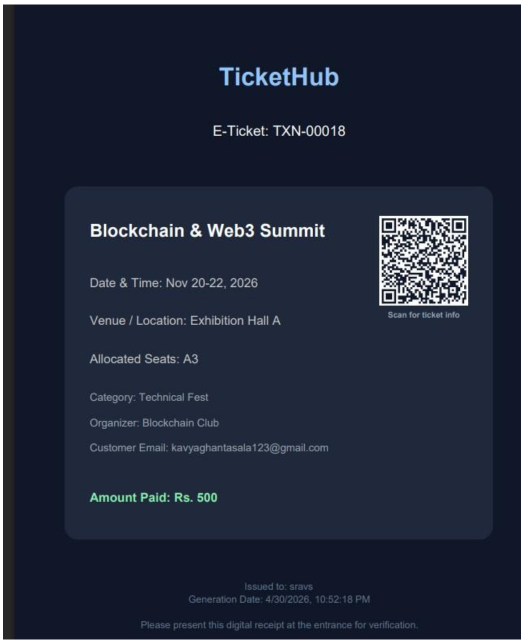

Figure 5.7: Ticket

5.3.8 Notification

The notification module (Fig. 5.8) informs users about booking confirmations and updates. It ensures timely communication by delivering alerts and important information related to their reservations.

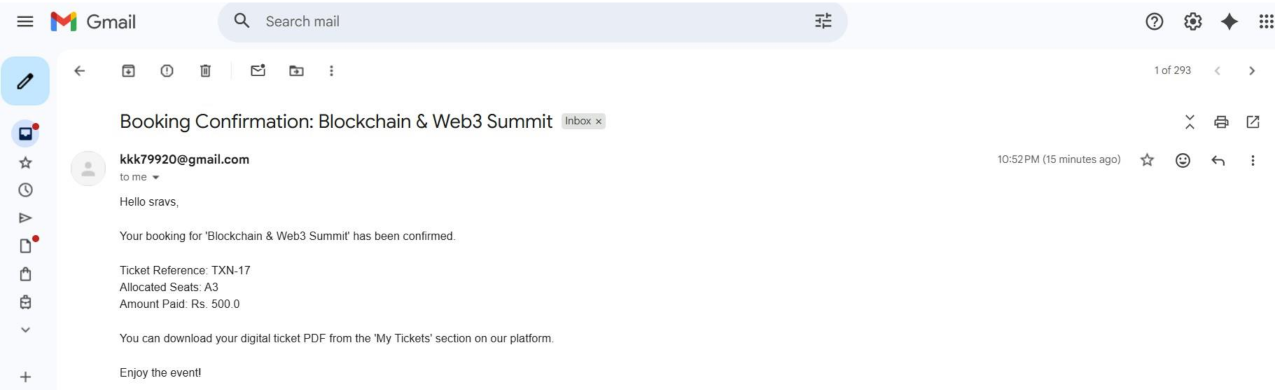

Figure 5.8: Notification

5.3.9 User Dashboard

The user dashboard (Fig. 5.9) provides a centralized view of all user activities, including booked tickets and event details.

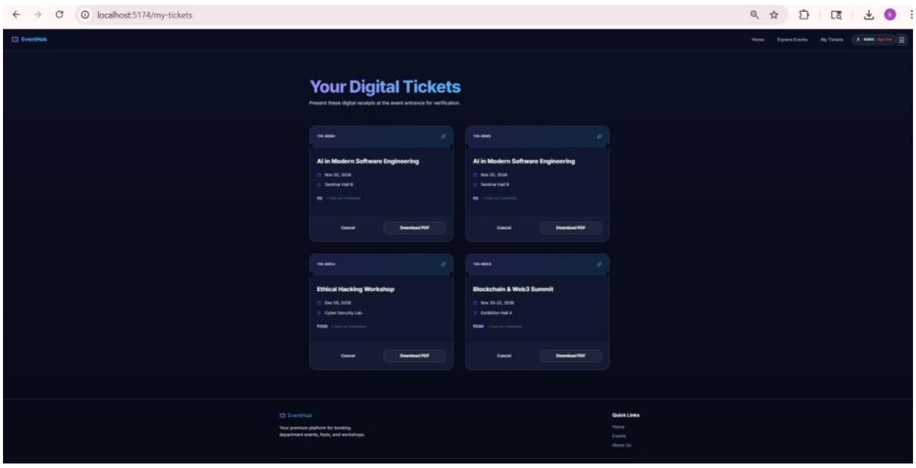

Figure 5.9: User Dashboard

5.3.10 Admin Management

The admin management module (Fig. 5.10) enables administrators to control system operations and manage users effectively.

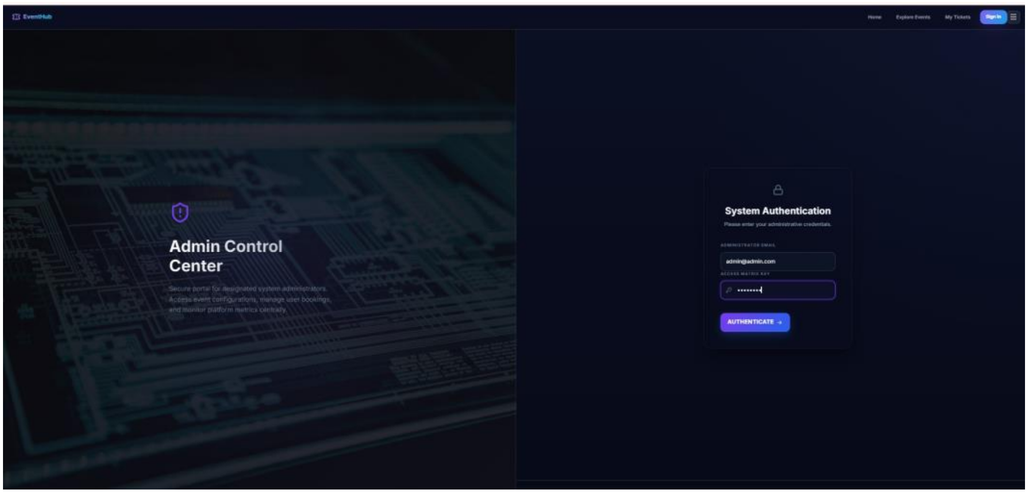

Figure 5.10: Admin Management

5.3.11 Admin Dashboard

The admin dashboard (Fig. 5.11) provides an overview of system data, helping administrators monitor activities and performance.

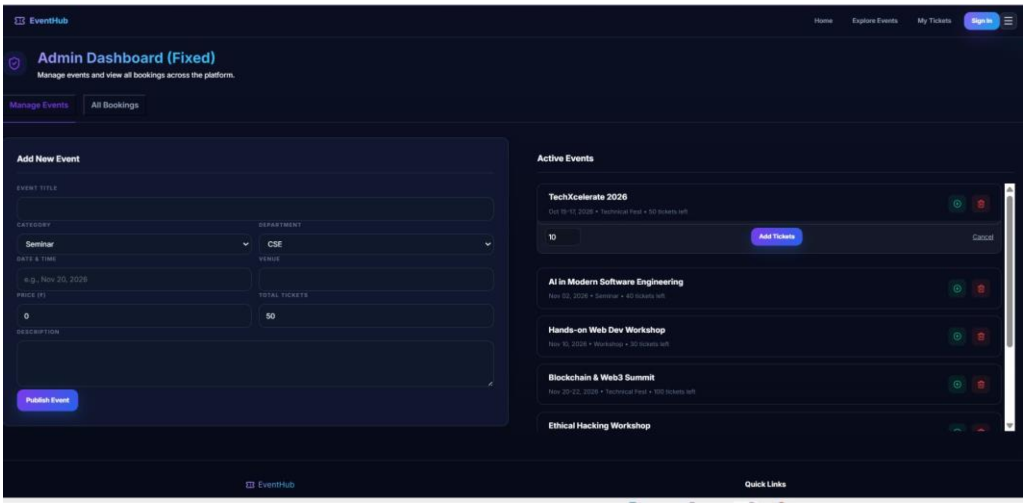

Figure 5.11: Admin Dashboard

5.3.12 Booking List

The booking list module (Fig. 5.12) displays all booking records with relevant details, allowing efficient tracking and management.

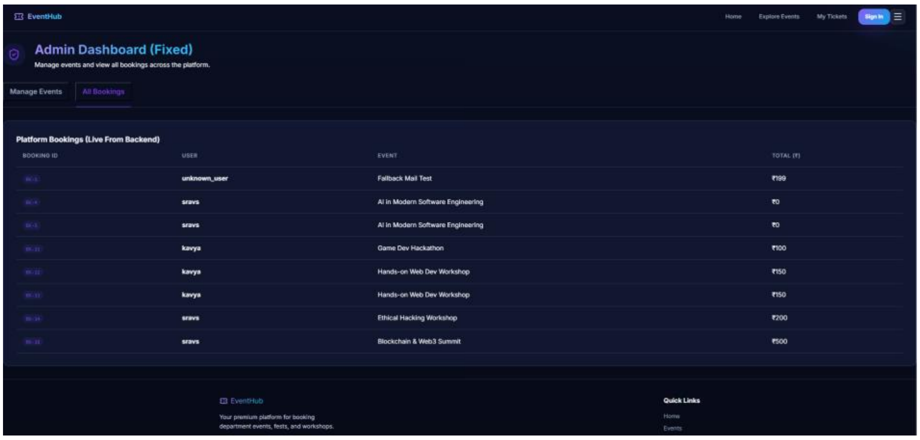

Figure 5.12: Booking List

5.3.13 Event Management Interface

The admin event management module (Fig. 5.13) enables administrators to manage active events effectively. It provides functionalities to add, update, and delete events from the system.

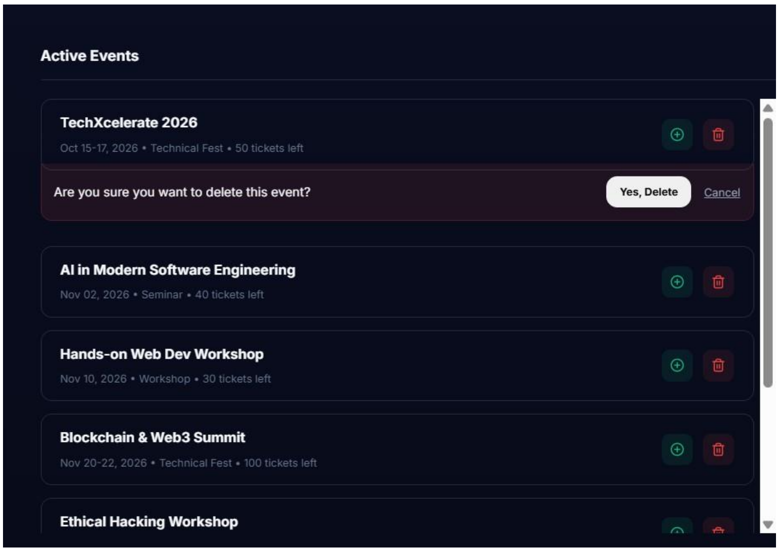

Figure 5.13: Event Management Interface

Chapter 6

CONCLUSION AND FUTURE ENHANCEMENTS

6.1 Conclusion

The Ticket Booking System for an Internal Department Event has been successfully designed and implemented to provide an efficient and reliable solution for managing event registrations. The system replaces traditional manual methods with a digital platform, thereby reducing time consumption, minimizing errors, and improving overall efficiency.

The application offers a user-friendly interface that allows users to easily browse events, check ticket availability, and complete bookings in a structured manner.

The integration of frontend, backend, and database components ensures smooth communication, real-time data updates, and secure handling of user information. The modular architecture of the system enhances scalability and makes it easier to maintain and upgrade in the future.

Overall, the system demonstrates how modern web technologies can be effec-tively utilized to simplify event management processes, improve user experience, and provide a reliable and organized solution for internal departmental activities.

6.2 Future Enhancements

While the current implementation provides a strong foundation, the following enhancements can further improve the system:

- **Online Payment Integration:** Integration of secure payment gateways to support digital transactions.
- **QR Code-Based Entry System:** Implementing QR code scanning for faster and secure event entry.
- **Mobile Application:** Developing a mobile app for easy access and real-time notifications.
- **Advanced Analytics Dashboard:** Providing insights on bookings, attendance, and event performance.
- **Email and SMS Notifications:** Sending automated alerts for confirmations and updates.
- **Role-Based Access Control:** Supporting multiple roles like Admin, Organizer, and Staff.
- **Seat Allocation Optimization:** Improving seat assignment for better user convenience.

Chapter 7

ADDRESSING COMPLEX ENGINEERING PROBLEM AND SDG

7.1 Mapping of the Project to Complex Engineering Problem (CEP)

Attribute	Characteristics	Justification
Depth of Knowledge	Requires in-depth engineering knowledge	Uses React JS, hooks, validation, and UI design concepts
Conflicting Requirements	Technical and non-technical constraints	Balances usability with booking rules and validation constraints
Depth of Analysis	Analytical and design thinking	Involves component design, state handling, and dynamic updates
Familiarity	Partially new problems	Real-time booking and ticket management features
Applicable Codes	Limited standard solutions	Requires custom booking and validation logic

Table 7. illustrates the key engineering characteristics involved in the development of the system. The project requires a strong depth of knowledge, as it involves modern frontend technologies like React JS, including hooks, form validation, and user interface design principles.

7.2 Mapping of the Project to Complex Engineering Activities (CEA)

Attribute	**Characteristics**	**Justification**
Range of Resources	Use of modern tools and technologies	React JS, HTML, CSS are used for development
Level of Interaction	Team and user interaction	Interaction between users and system through UI
Innovation	Use of creative solutions	Custom ticket booking and validation features
Consequences	Impact on users	Ensures accurate booking and user satisfaction

Table 7.2 illustrates how the project aligns with Complex Engineering Activities (CEA). The system makes use of a wide range of resources, including modern web technologies such as React JS, HTML, and CSS, which support efficient and scalable application development.

7.3 Sustainable Development Goals (SDG) Mapping

SDG Goal	**Description**	**Project Contribution**
Quality Education (SDG 4)	Ensure inclusive and equitable education	Helps students easily register for events and activities

Industry Innovation (SDG 9)	Build resilient infrastructure	Promotes digital solutions for event management
Sustainable Cities (SDG 11)	Smart and inclusive systems	Encourages paperless ticket booking system

Table 7.3 presents the alignment of the project with relevant Sustainable Development Goals (SDGs). The system contributes to **Quality Education (SDG 4)** by providing a platform that enables students to easily register for events, workshops, and academic activities, thereby improving accessibility and participation.

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[8] MySQL Documentation: https://www.mysql.com

[9] Axios Documentation: https://axios-http.com